

SAI GANGADHAR POTLURI

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TECHNICAL SUMMARY

Electrical Engineering graduate from IIT Bhubaneswar with a strong analytical foundation, transitioning into software engineering. Proven capability in architecture-driven Python development, relational database modeling, and building stateful backend applications. Experienced in optimizing operational workflows and business pipelines using systematic data handling and metric analysis.

EDUCATION

Indian Institute of Technology (IIT) Bhubaneswar
Bachelor of Technology (B.Tech.) in Electrical Engineering

TECHNICAL SKILLS

- Programming Languages: Python 3, SQL (SQLite, PostgreSQL)
- Backend & Web Frameworks: Flask, REST APIs, Object-Oriented Programming (OOP)
- Data Analytics: Pandas, NumPy, Data Cleaning, Manipulation & Advanced Visualization
- Developer Tools: Git, GitHub Version Control, Linux Shell, JSON Data Serialization, VS Code
- Core Concepts: Data Structures & Algorithms (DSA), Relational Database Management Systems (DBMS)

SOFTWARE ENGINEERING PROJECTS

Full-Stack Personal Finance & Analytics Ledger (Flask & SQLite)

- Engineered and deployed a multi-user, stateful web application tracking data transaction arrays.
- Implemented secure user session state controls utilizing cryptographic security hashing protocols.
- Modeled relational SQLite schemas to store operational transactional entries with localized ZoneInfo timestamping.
- Developed custom data filtering structures and data streaming pipelines for on-demand CSV report generation.
- Technologies Used: Python, Flask, SQLite3, HTML5/CSS3, Python ZoneInfo, Render Cloud Engine

Flashcard & Spaced-Repetition Studio (Core Python Object-Oriented Architecture)

- Designed and built an algorithmic command-line active recall compiler built on robust OOP principles.

- Implemented a custom programmatic version of the Leitner Spaced-Repetition System to graduate user data mastery boxes based on dynamic performance evaluation thresholds.
- Integrated structured JSON file serialization protocols to persist tracking matrices and performance history across application runtimes seamlessly.
- Technologies Used: Pure Python 3, OOP Principles, JSON File Handling, Custom Algorithm Logic

Individual Capstone Project: Solar Tracker Irradiance Optimizer (MATLAB & Simulink)

- Independently designed, modeled, and simulated a physical solar energy extraction optimizer utilizing the Perturb and Observe (P&O) Maximum Power Point Tracking (MPPT) algorithmic matrix.
- Programmed control loop feedback systems completely from scratch to evaluate photovoltaic cell efficiency responses under fluctuating atmospheric irradiance data.
- Handled the end-to-end engineering pipeline, from algorithmic mathematical derivation to system validation inside MATLAB/Simulink.

PROFESSIONAL EXPERIENCE

UNext Learning | March 2025 – January 2026

Data-Driven Education & Operations Consultant

- Managed and audited customer onboarding databases, using organized data tracking models to eliminate workflow redundancies.
- Utilized data-driven requirement analyses on lead pipelines to segment target audiences, optimizing operational efficiency.
- Maintained data tracking metrics to monitor conversion trends, reporting key performance updates to stakeholders.

Unacademy | March 2024 – July 2024

Business Development Associate

- Structured, systematic communication and data tracking protocols to manage high-volume customer relationship matrices.
- Handled end-to-end user onboarding pipelines, analyzing learner requirement metrics to design optimized educational pathways.
- Conducted data-backed consultative requirement mapping to increase client onboarding efficiency systematically.

Hygiene Big Bites | October 2023 – February 2024

Kitchen Operations & Logistics Manager

- Oversaw daily inventory management datasets, optimizing supply chain loops to reduce resource waste.
- Tracked operational metrics and inventory depletion cycles to identify logistical bottlenecks, cutting processing overhead.

- Coordinated cross-functional activities using structured scheduling systems to maximize kitchen throughput.

OPTCL (B.Tech Internship)

Power Systems Analysis Intern

- Analyzed power grid distribution network datasets to evaluate system loading states and calculation thresholds.

LEADERSHIP & CAMPUS ACHIEVEMENTS

- Core Team Member – E-Cell (Entrepreneurship Cell), IIT Bhubaneswar: Formulated operational strategies, coordinated campus-wide initiatives, and drove strategic leadership frameworks to cultivate startup ecosystems.
- Varsity Basketball Player – IIT Bhubaneswar (2016 – 2018): Selected for the competitive institute team; maintained a rigorous training pipeline emphasizing team synergy, tactical execution, and peak high-pressure execution.
- Portfolio Verified Developer: Architected and live-deployed production-ready web codebases and algorithmic command-line interfaces visible via public GitHub tracking.